Algebra 1
Unit 8
Lesson 8 Homework

Name Answer Key ____________
Date____________________
Period___________________

In 1965, a Ford Mustang cost \$2,600. While most cars lose value, a 1965 Mustang in immaculate condition can be worth over \$30,000. The function $M(x) = 20x^2 - 420x + 2600$ models the price of the 1965 Mustang in the United States since 1965.

1. What does the variable x represent? Include the units.
   Years since 1965

2. What does the variable y represent? Include the units.
   Value of the Mustang, in dollars

3. Interpret the y-intercept. Include units.
   (0,2600) In 1965, the Mustang was worth \$2600

4. Rewrite the function in vertex form. Show your work.
   $M(x) = 20(x^2 - 21x + 110.25) + 2600 - 2205$
   $M(x) = 20(x - 10.5)^2 + 395$

5. Interpret the vertex in terms of the context. Include units.
   (10.5,395)
   The Mustang's lowest value was \$395, 10.5 years after 1965.

6. Complete the table of values for $M(x)$.

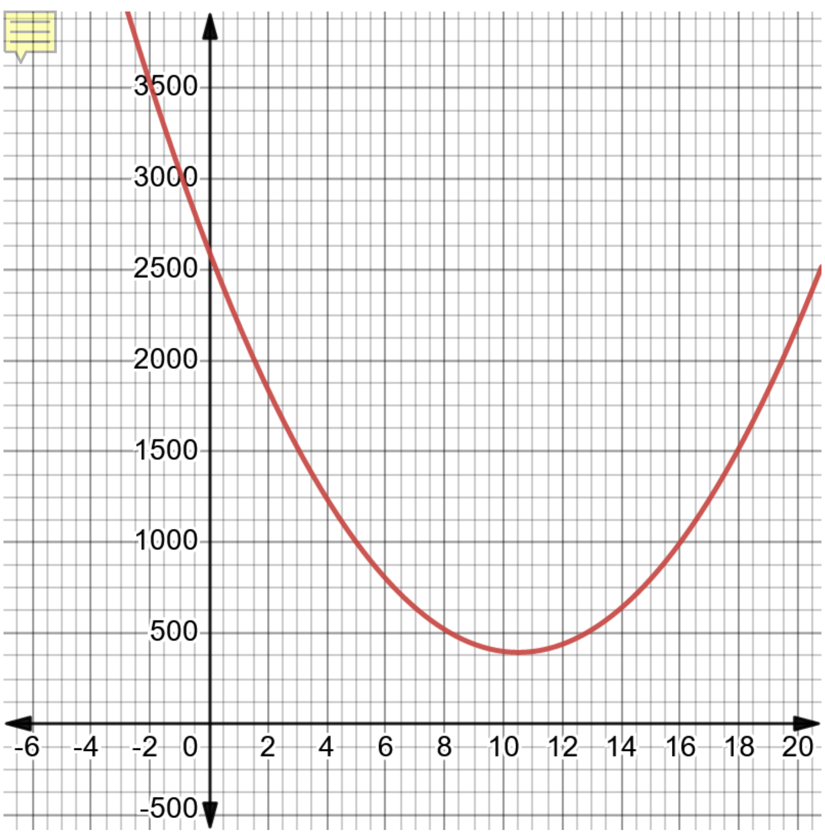
x	-5	-3	0	7	10.5	12	24	50
$M(x)$	5200	4040	2600	640	395	440	4040	31600

7. Circle the values in the table that do not make sense in terms of the context.

8. Complete as many of the key features as possible using the given function, the function rewritten in vertex form, and the table of values. Do not use approximate or rounded values.

Key Features					
x-intercept		vertex	$(10.5,395)$	domain	All Real
x-intercept		increasing	$x > 10.5$	range	$y \geq 395$
y-intercept	$(0,2600)$	decreasing	$x < 10.5$	opens	Up
left end behavior		$x \to -\infty$, $y \to$ $\underline{\infty}$			
right end behavior		$x \to \infty$, $y \to$ $\underline{\infty}$			

9. Graph the function.



10. Use the graph, table of values, or function to determine the reasonable constraints for the context. Write the constraints in set builder notation.

$\{x | 0 \leq x \leq 60\}$ $\{y | y \geq 395\}$